



Storage Park

We're all looking for that sweet spot in storage devices where we get the best of both the worlds of capacity and performance. Well, this hard disk drive shootout will seek out just that!

Today, upgrading a hard drive has almost become a fashion statement—gone are the days when an 8 GB hard disk was considered sufficient. Everything from MP3 songs, DivX movies, 3D games, applications and even operating systems place tremendous demands on hard disk space. These days if you own anything below 20 GB you are sure to run into that roadblock of a 'disk full' message!

Along with just space requirements, there are increasing demands on the performance of drives as well. Multimedia applications (video and audio) place increasing demands on bandwidth. Besides, today's operating systems allow smoother and increased levels of multitasking, allowing you to work with several applications simultaneously—and hard disks have to be able to keep up with loading and running these demanding applications efficiently.

In keeping with all this, there has been a tremendous increase in the performance and reliability of hard disks today. Buying a 7,200-rpm 80 GB drive now certainly makes a value proposition, whereas just recently most would have considered a 40 GB 5,400-rpm drive a great value buy.

Let's take a look at where things stand today. In this comparison, we compared the best of the crop of hard disks over two categories—let's see which ones turn out top dog.

Digit Test Process

The hard drives we categorised into two groups: those offering up to 60 GB of storage space and those with more than 60 GB capacity.

The test bed comprised a Pentium 4 running at 2.8 GHz on an Intel D850EMV2 board with 512 MB of RDRAM. The graphics card used was a GeForce 3 with 64 MB of DDR RAM and a 40 GB Seagate ATA/4 hard disk, which was installed on the primary IDE channel. All the test drives were connected to the secondary channel as a master. The necessary drivers, nVidia Detonator version 28.32 for the graphics card and the Intel motherboard's INF Update version 4.00.1009 were loaded after installing Windows XP Professional as the operating system. The benchmarking utilities were loaded on to the drive and all drives were verified to be working in DMA mode before running the test.

Test methodology

We evaluated the hard disks for features (10 per cent weightage), performance (50 per cent), warranty and support (20 per cent) and value for money (20 per cent). These weightages formed the

basis for calculating the overall score for each product.

Features: As part of the overall scoring of the hard disks, we looked for different features of the drive such as the spindle speed, data buffer size, type of interface and rated access time. We also considered the documentation (manuals and quick start guides) and accessories such as data cables, mounting plates and screws.

Performance: We used the following benchmarks and applications to evaluate the performance of the hard disks:

HDTach 2.6: This is a synthetic test in which we recorded the read burst speed and CPU utilisation. Here, data is read from the disk and then written back to the disk. A graph is displayed as the data is read and written in real time. During this test, the program calculates the total CPU usage of the drive and the burst transfer rate of the drive under test. The burst transfer rate is a measure of the maximum possible bandwidth of the drive over its interface.

SiSoft Sandra 2002 Pro: This is also a synthetic benchmark. It evaluates the sequential and random read and write transfer rates and the access time of the disk. The benchmarking suite continuously writes and reads the data to and from the disk and measures these parameters using different data sets. We also recorded the SiSoft Sandra 2002 index, which is an aggregation of the performance of the drive over these parameters.

Return to Castle Wolfenstein: This real world test is a good indicator of how a hard disk performs under heavy stress when running one of the newer and demanding games. We loaded the retail version of the game *Return to Castle Wolfenstein* from the test hard drive and noted down the time taken to load the first map.

Photoshop 7: We installed Photoshop 7 on the test hard drive and a 200 MB TIFF image was copied on to it. We first recorded the time taken for Photoshop 7 to load, which was part of the application load up time test. The TIFF image was then opened from the test hard drive and the time taken for Photoshop 7 to open this file was recorded. This test is a good indicator of random and sequential read speeds using a real world application.

Warranty and support: Here we took into account the number of years of replacement warranty provided by the manufacturer and the extent of the service centre network.

Value for money: While buying any product, value for money

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In 1956, IBM introduced the first storage system that could store a maximum of 5 MB of data and it featured disks with 24-inch diameters!